

Colin D. Gray

336-471-9817 | cdgray@caltech.edu | Portfolio: <https://colingray3149.github.io> →



EDUCATION

California Institute of Technology (Caltech) – Chemical Engineering (Process Systems), Pasadena CA

Jun 2027

Overall GPA: 3.8/4.00

Relevant Coursework: Organic Chemistry, ChemE Thermodynamics, Python for ChemE, Transport Phenomena, Chemical Kinetics, Separations and Distillations, Differential Equations, Partial Differential Equations

EXPERIENCE

Independent Researcher (Self-Directed) – Machine Learning/Computational Combustion

Advised by Prof. Tarek Echekki (NC State University) - *Remote*

May 2025 – Sep 2025

- Developed a framework to accelerate and simplify combustion simulations by pruning redundant chemical species
- Built a novel reproducible Python/Cantera pipeline to simulate methane-air and gasoline surrogate combustion
- Generated and processed over 60 gigabytes of data regarding conditions across 400K ranges of temperatures and 0.5-40 atm
- Trained PyTorch regression models using LASSO and “gated heads” to reduce species in combustion mechanisms by up to 80%
- Automated evaluation of reduced mechanism performance, noting simulation speed increases of up to 400% and 90% accuracy

Propulsion Team Member

Propulsion, Aeronautics, and Rocket Systems Engineering at Caltech (PARSEC) – *Pasadena CA*

Sep 2025 – Present

- Collaborating with a multidisciplinary team to fire a liquid-fueled rocket capable of reaching 45,000 ft and recovering intact
- Lead data acquisition for the Rudy RP-1/LOX rocket engine – integrating and programming a full sensor suite
- Measuring convective heat flux from the combustion gases with thermocouples embedded in combustion chamber walls
- Determining thermodynamic state of propellant liquids prior to injection into combustion chamber with pressure transducers

Summer Research Fellow – Jet Fuel Autoignition

Explosion Dynamics Laboratory - Caltech - *Pasadena CA*

Jun 2024 – Aug 2024

- Supporting the development of sustainable aviation fuels by verifying ignition correlations through experimentation
- Used Excel/MATLAB to investigate autoignition temperature (AIT) correlations in jet fuels and identified 7 paths to research
- Planned and conducted experiments verifying correlations between branches and compactness of molecules to AIT
- Analyzed industry standard and surrogate fuels using GC-MS data and AIChE DIPPR chemical database
- Presented data at two conferences, a seminar, weekly meetings with the lab group, and monthly meetings with Boeing

Teaching Assistant

Caltech - *Pasadena CA*

Jan 2024 – Present

- Head TA – SA16C Cooking Basics
- TA – ChE63a Chemical Engineering Thermodynamics
- TA – Ge01 Earth and Environment

Summer Intern – Drug Discovery Research

Chen Group - Wake Forest School of Medicine - *Winston-Salem NC*

Jun 2022 – Aug 2022

- Furthered the development of drugs designed to lessen the effects of cocaine addiction and withdrawal
- Researched cholesterol regulation of glutamine receptors (mGluRs) using multiple biochemical assays
- Imaged cells using advanced microscopy (TIRF, confocal microscopy) and processed images using ImageJ and Excel

INDEPENDENT PROJECTS

2D Cold-flow Scramjet - Ansys Fluent/Workbench (Paused/In Progress)

Aug 2025 – Sep 2025

- Simulating supersonic airflow through a scramjet inlet to understand mixing behavior prior to combustion

Compartment/Dresser Fire - Pyrosim/FDS + Smokeview

Jun 2025 – Aug 2025

- Modeling real-scale room fire dynamics to study heat transfer, smoke propagation, and ventilation effects using CFD

VOLUNTEERING

Emergency Medical Responder (EMR) Certified Volunteer

American Red Cross - *Los Angeles CA*

Jun 2025 – Present

- Provided support to 100+ members of the community at several events, including the Rose Parade and LA Pride Parade

SKILLS

- | | |
|---|---|
| • Coding: Python (NumPy, Pandas, PyTorch, Cantera) | • Microsoft Suite (Excel, Word, PowerPoint) |
| • Chemical Processing Plant Design (DWSIM) | • Computational Fluid Dynamics (FDS, Ansys, COMSOL) |
| • Scientific Presenting (at Seven Conferences/Seminars) | • Community Service and Engagement |